BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 017 · Defensive Design, Validation and Testing

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does assignment do to a variable?

Q2. What does the MDR hold?
A. Data moving to/from memory
B. The next instruction's address
C. Only ALU calculation results
D. Only user account names

Q3. What does the Program Counter hold?

Q4. Similar CPUs have different clock speeds and cores. Some tasks can be shared across cores. Correct both claims. Explain why. (a) Double clock speed always halves run time. (b) Each extra core doubles every program's speed.

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I Do · Read new knowledge

Key vocabulary

defensive design — Designing programs to handle invalid input and misuse.

validation — Checking entered data against defined rules.

authentication — Checking a user's identity.

boundary data — Valid data at the extreme limits of an allowed range.

New learning

Robust programs: Defensive design anticipates invalid or malicious input and uses readable, maintainable code.

Validation: Validation checks input against rules such as type, length or range. It does not verify a person's identity.

Authentication: Authentication establishes identity, for example using credentials or another factor.

Testing: Use normal, boundary, invalid and erroneous values; compare expected and actual results. Test during development and again at the end.

Worked example

A program accepts a mark from 0 to 100 inclusive. Choose test data and explain the expected results.

1. Write the validation rule: 0 <= mark <= 100.

2. Choose normal data: 56 → accepted.

3. Choose boundary data: 0 and 100 → accepted.

4. Choose invalid data of the correct type: -1 or 101 → rejected.

5. Choose erroneous data of the wrong type, e.g. 'hello' → rejected or handled safely.

We Do · Practise with guidance

Guided 1

Classify input 50 for a 0–100 mark rule and state expected outcome.

Worked method: Normal valid data; accepted.

Guided 2

Classify 100 and 101.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: 100 is valid upper boundary and accepted; 101 is invalid and rejected.

Guided 3

Why can a valid username format fail authentication?

Worked method: Validation checks format; authentication checks identity/credentials and can still fail.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the number printed when one is added.

```
value = 255
print(value + 1)
```

predict

Question 1

Define input validation. [2 marks]

Question 2

Give two boundary values and one invalid value for marks from 0 to 100 inclusive. [3 marks]

Question 3

Distinguish authentication and validation; explain why iterative testing is useful. [4 marks]

Exit Ticket · Answer before checking

Validation and authentication mean exactly the same thing.

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It stores a value in the named variable, replacing its previous value if one existed.

Q2. A — Data moving to/from memory

Q3. The address of the next instruction to be fetched.

Q4. (a) Higher clock speed means more cycles per second, which may process more instructions. (b) More cores help when work can be shared between them. Speed also depends on the task and other CPU features.

I Do / We Do pages · Guided practice

Q1. Normal valid data; accepted.

Q2. 100 is valid upper boundary and accepted; 101 is invalid and rejected.

Q3. Validation checks format; authentication checks identity/credentials and can still fail.

You Do pages · Code First

```
value = 255
print(value + 1)
```

Expected result: 256

You Do pages · Independent questions

Q1. Checking input against specified rules [1], rejecting unsuitable data before processing [1].

Q2. Boundary 0 [1] and 100 [1]; invalid e.g. 101 or -1 [1].

Q3. Authentication checks identity [1]; validation checks data against rules [1]; iterative testing happens during development [1], identifying and correcting faults early [1].

Exit Ticket · Model answer

False

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: